Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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***Abstract*—An unmanned surface vessel (USV) and an unmanned aerial vehicle (UAV) connected by a tether combine the endurance of the USV with the view from above of the UAV, which draws power and a reliable communication link through the tether. The tether, however, couples the two vehicles, which must remain within its reach while each respects its own limits, under waves and wind. This work develops the cooperative control of such a system, formed by the WAM-V USV and the x500 UAV. Models of both vehicles are derived and validated against a simulator, together with a closed-form model of the force of a slack tether that accounts for its weight. Each vehicle is driven by its own non-linear model predictive controller, and the two controllers exchange their predicted trajectories, so that each anticipates the motion of the other within the reach of the tether. A disturbance observer compensates unmodelled forces, such as the wind. In a simulation at sea, the UAV controller tracks a trajectory with less than a quarter of the error of the PX4 position controller, and both controllers keep their vehicles within their limits when the reference cannot be followed. The UAV lands on the moving deck a few centimetres from the centre of the pad in waves of up to $1.2\,$m and winds of up to $8\,$m/s, and in missions of tower inspection, target pursuit and UAV recovery, the two vehicles operate as a single system.**

***Index Terms*—Cooperative control, non-linear model predictive control, tether, unmanned aerial vehicle, unmanned surface vessel**

I. Introduction

SAVING lives at sea, protecting marine ecosystems and securing maritime infrastructure all rely on a continuous and local awareness of what happens at sea [1]. Unmanned surface vessels (USVs) keep a presence at sea for long periods at a low cost, but they only see as far as the horizon, while unmanned aerial vehicles (UAVs) sense from above, but have a short endurance and an unreliable long-range link [2], [3]. A tether between the two supplies the UAV with power and a reliable communication link from the USV, which removes both limits at once [3], and such systems are already offered by industry [4], [5].

The tether, however, turns the two vehicles into a single system. The distance between them must stay within the reach of the tether, so that the motion of one restricts the motion of the other, and the two move on very different time scales, the USV slowly, dominated by its inertia and by the drag of the water, and the UAV quickly. Wind and waves disturb both vehicles, and the UAV must take off from and land on a deck that moves with the waves. Operating the pair as a single system therefore requires both a model of the tether that accounts for its weight and a coordination in which each vehicle anticipates the motion of the other.

This work addresses the control of a USV and a UAV connected by a tether, so that they carry out maritime missions autonomously as a single system. Each vehicle is driven by its own non-linear model predictive controller (NMPC), and the two controllers exchange the trajectories they predict, so that each keeps the separation within the reach of the tether by taking into account where the other vehicle is going, rather than where it is. The controller of the UAV also predicts the force of the hanging tether, whose slack is kept constant by a winch on the deck of the USV, and a disturbance observer compensates the forces that the models do not describe, such as the wind. Simple manoeuvres, such as a takeoff, an orbit or a landing, give the reference of each vehicle, and missions chain them into complete operations. The main contributions are a closed-form model of the force of a slack tether that accounts for its weight, the two cooperating NMPCs, manoeuvres for the takeoff from and the landing on the moving deck and for the inspection of a structure by both vehicles, and a simulation of the tethered system at sea in which they are evaluated.

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II. Related Work

Tether models. A heavy tether suspended between two points takes the shape of a catenary, whose sag and tension follow from the equations of static equilibrium [6]. Such quasistatic models are inexpensive, but neglect the dynamics of the tether, which lumped-mass models capture by discretising it into a chain of nodes joined by spring–damper segments, as in MoorDyn [7], at a higher computational cost. In tethered and cooperative UAV systems, the tether is often simplified further, either modelled as massless and always taut, so that it pulls along the straight line between its ends, as in the buoy–UAV system of Kourani and Daher [8] and the cooperative manipulation of Novák et al. [9], or treated as an unknown disturbance, estimated online and rejected by the controller [10]. Neither describes a slack tether, whose weight turns its force away from that straight line.

Control and cooperation. Predictive control is a common choice for UAV–USV systems, since it can state the coupling between the vehicles as constraints over a prediction horizon. Li et al. [11] use an NMPC to drive a UAV to track and land on a moving USV, but only the UAV is controlled, and the USV moves independently. Zhang et al. [12] adopt a leader–follower structure, in which the USV follows a predefined path and the UAV keeps a desired position relative to it with its own NMPC. Novák et al. [9] compute the commands of both

vehicles in a single, centralised problem, with tethers modelled as massless and taut. In each case, either the tether is simplified or the coordination runs in one direction only, with one vehicle adapting to the other without being anticipated by it.

III. Modelling and Control

Each controller predicts the motion of its vehicle over a short horizon, so its model must be accurate enough to anticipate that motion, yet simple enough to be evaluated many times within each sampling interval. The models are derived from the physics of each vehicle, and the force of the tether is described by a closed-form model that accounts for its weight.

A. USV Model

The USV, a catamaran, moves on the surface, so its controller only needs to predict its motion in the horizontal plane. The state collects its position (x, y) and heading ψ in the inertial frame, and its surge, sway and yaw velocities (u, v, r) in its body frame. The kinematics are $\dot{x} = u\cos\psi - v\sin\psi$, $\dot{y} = u\sin\psi + v\cos\psi$ and $\dot{\psi} = r$, and the dynamics follow from Newton's and Euler's laws in the rotating body frame, with the linear-plus-quadratic hydrodynamic damping of [13],

$$\begin{aligned} \dot{u} &= \tfrac{1}{m}\big[X - (X_u + X^{\pm}_{uu}|u|)u\big] + vr, \\ \dot{v} &= \tfrac{1}{m}\big[Y - (Y_v + Y_{vv}|v|)v\big] - ur, \\ \dot{r} &= \tfrac{1}{I_z}\big[N - (N_r + N_{rr}|r|)r\big], \end{aligned} \tag{1}$$

where m and I_z are the mass and the yaw inertia, and the quadratic surge damping $X^{\pm}_{uu}$ may differ between forward and backward motion. The force and moment (X, Y, N) are produced by two azimuth thrusters, with thrusts T_L, T_R and angles α_L, α_R, mounted behind the centre of gravity at $(-x_{arm}, \pm y_{arm})$,

$$\begin{aligned} X &= T_L\cos\alpha_L + T_R\cos\alpha_R, \\ Y &= T_L\sin\alpha_L + T_R\sin\alpha_R, \\ N &= -x_{arm}(T_L\sin\alpha_L + T_R\sin\alpha_R) \\ &\quad - y_{arm}(T_L\cos\alpha_L - T_R\cos\alpha_R). \end{aligned} \tag{2}$$

B. UAV Model

The UAV, a quadrotor, is described by its position $\mathbf{p} = [x, y, z]^\top$ and velocity in the inertial frame. Its acceleration results from the thrust T along the vertical axis of its body, $\mathbf{z}^{\mathcal{B}}$, gravity, a linear rotor drag d_r on the horizontal velocity, and the force of the tether,

$$\ddot{\mathbf{p}} = \frac{T}{m}\,\mathbf{z}^{\mathcal{B}} - \begin{bmatrix}0\\0\\g\end{bmatrix} - \frac{d_r}{m}\begin{bmatrix}\dot{x}\\\dot{y}\\0\end{bmatrix} + \frac{1}{m}\mathbf{F}_{\text{tether}}. \tag{3}$$

The controller does not drive the rotors, but sends attitude and thrust setpoints to the autopilot of the UAV, PX4 [14], which tracks them with its own attitude controller. The attitude therefore follows its command with a delay. The autopilot rotates the UAV so that the direction of the thrust approaches the commanded one, and controls the yaw separately, which is described by two first-order responses, with time constants τ and τ_ψ, to commands received after transport delays τ_d and $\tau_{d,\psi}$,

$$\begin{aligned} \dot{\mathbf{z}}^{\mathcal{B}} &= \tfrac{1}{\tau}\big(\mathbf{R}(\phi_{\text{cmd}}, \theta_{\text{cmd}}, \psi_{\text{cmd}})\big|_{t-\tau_d}[0,0,1]^\top - \mathbf{z}^{\mathcal{B}}\big), \\ \dot{\psi} &= \tfrac{1}{\tau_\psi}\big(\psi_{\text{cmd}}(t-\tau_{d,\psi}) - \psi\big), \end{aligned} \tag{4}$$

where $\mathbf{R}$ is the rotation of the body for the commanded roll, pitch and yaw. Since the delayed commands are not part of the state, they are approximated to first order by the commands and their rates, $\phi_{\text{cmd}}(t-\tau_d) \approx \phi_{\text{cmd}} - \tau_d\,\dot{\phi}_{\text{cmd}}$, and likewise for the pitch and the yaw.

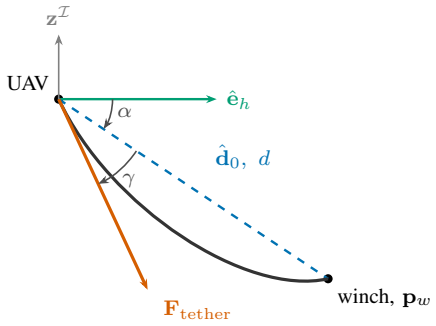


Fig. 1. Force of the tether on the UAV, in the vertical plane of the catenary.

C. Tether Force Model

The tether runs from the UAV to a winch on the deck of the USV, at $\mathbf{p}_w = [x_v, y_v, z_t]^\top$, above the centre of gravity of the USV at a fixed height z_t. The winch keeps the length L of the tether slightly longer than the distance $d = \|\mathbf{r}\|$ between its ends, $\mathbf{r} = \mathbf{p}_w - \mathbf{p}$, so that its slack ratio $\varepsilon_0 = (L-d)/L$ stays constant. A tether of linear mass μ then hangs as a catenary, and the tension at the UAV is close to the weight of the tether, $\mu g L$, so that it grows in proportion to the distance,

$$T_{\text{tether}} = c_d\,d, \qquad c_d = \frac{\mu g}{1-\varepsilon_0}. \tag{5}$$

The weight of the tether also turns its force away from the chord, the straight line $\hat{\mathbf{d}}_0 = \mathbf{r}/d$ towards the winch, by an angle γ towards the vertical, as illustrated in Fig. 1. Only the weight normal to the chord bends the tether, so the deflection is largest for a horizontal chord, where it equals the angle γ_0 of the catenary at its end, and vanishes for a vertical one. It is therefore taken as $\gamma = \gamma_0\cos\alpha$, where α is the depression angle of the chord below the horizontal direction $\hat{\mathbf{e}}_h$ towards the winch, and the force on the UAV is

$$\mathbf{F}_{\text{tether}} = c_d\,d\,\big[\cos(\alpha+\gamma)\,\hat{\mathbf{e}}_h - \sin(\alpha+\gamma)\,\mathbf{z}^{\mathcal{I}}\big], \tag{6}$$

a closed-form expression of the positions of the two vehicles, which replaces the numerical solution of the catenary at every step of the prediction.

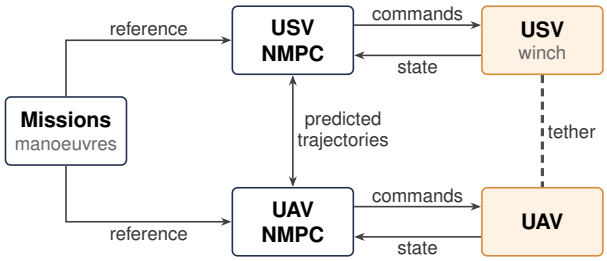


Fig. 2. Overview of the system.

D. Cooperative NMPC

Each vehicle is driven by its own NMPC, which solves, at every sampling instant k, an optimal control problem over a horizon of N steps,

$$\begin{aligned}\min_{\hat{\mathbf{x}},\hat{\mathbf{u}},\hat{\boldsymbol{\sigma}}} \; & \sum_{i=0}^{N-1} \ell(\hat{\mathbf{x}}_i,\hat{\mathbf{u}}_i) + \ell_f(\hat{\mathbf{x}}_N) + \sum_{i,j}\rho(\hat{\sigma}_{j,i}) \\ \text{s.t. } & \hat{\mathbf{x}}_0 = \mathbf{x}_k, \quad \hat{\mathbf{x}}_{i+1} = \boldsymbol{\Phi}(\hat{\mathbf{x}}_i,\hat{\mathbf{u}}_i), \\ & \hat{\mathbf{x}}_i \in \mathcal{X}, \quad \hat{\mathbf{u}}_i \in \mathcal{U}, \quad h_j(\hat{\mathbf{x}}_i) \le \hat{\sigma}_{j,i}, \quad \hat{\sigma}_{j,i} \ge 0,\end{aligned} \tag{7}$$

where $\boldsymbol{\Phi}$ integrates the model over one sampling interval, the stage cost ℓ and the terminal cost ℓ_f penalise the deviation from the reference, and only the first input is applied before the problem is solved again. The limits that the vehicle cannot exceed, $\mathcal{X}$ and $\mathcal{U}$, are hard, whereas the constraints h_j on states that a disturbance can push beyond their bounds are softened by slacks σ_j, penalised by $\rho(\sigma) = \mu_1\sigma + \mu_2\sigma^2$, so that the problem always has a solution and a constraint is violated only when it cannot be met. Since the constraints act on the predicted trajectory, the controller keeps the vehicle within its limits even when the reference itself cannot be followed, so that the reference can be as simple as a single point.

A single controller for both vehicles would impose one horizon and one sampling time on vehicles that move on very different time scales, and enlarge the problem to be solved within each sampling interval. Each vehicle therefore has its own NMPC, and the two exchange the trajectories they predict, as shown in Fig. 2. Each controller resamples the latest prediction of the other at the nodes of its own horizon, and constrains the separation between the vehicles along it, so that it anticipates where the other vehicle is going. The horizon of the UAV is shorter than that of the USV, so the USV enforces this constraint only over the part of its horizon that the prediction of the UAV covers, a window that slides forward as the problem is solved again at every instant.

USV NMPC. To bound both the thrusts and how fast they change, the thrusts and angles become part of the state, $\mathbf{x}_v = [x, y, \psi, u, v, r, T_L, T_R, \alpha_L, \alpha_R]^\top$, and the inputs are their rates, so that their ranges act on the state and their rate limits on the input. The stage cost penalises the errors of position, heading and velocity, together with the thrusts and their rates. A soft lower bound on the surge speed keeps the USV sailing bow first, and the separation from the predicted trajectory of the UAV, $\hat{\mathbf{p}}_a$, is limited to the reach of the tether, d_{max},

$$(x-\hat{x}_a)^2 + (y-\hat{y}_a)^2 + (\hat{z}_a - z_t)^2 - d_{max}^2 \le \sigma. \tag{8}$$

UAV NMPC. In the same way, the commanded attitude and thrust become part of the state, and the inputs $\mathbf{u}_a$ are their rates. The prediction model combines (3)–(6), with the winch placed on the predicted trajectory of the USV. The stage cost

$$\begin{aligned}\ell = & \|\mathbf{p}-\mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v}-\mathbf{v}_{ref}\|_{Q_v}^2 + \left\|[\phi,\theta]^\top\right\|_{Q_{tilt}}^2 \\ & + q_\psi\big(1 - w_b\cos(\psi-\psi_{ref})\big) \\ & + Q_T\,(T-mg)^2 + \|\mathbf{u}_a\|_R^2\end{aligned} \tag{9}$$

penalises the errors of position and velocity, the tilt, the deviation of the thrust from its hover value and the rates of the commands. It also turns the UAV towards a point of interest, such as the winch, a structure or a target, whose bearing is ψ_{ref}, with a weight $w_b = \|\mathbf{b}\|/\sqrt{\|\mathbf{b}\|^2+\delta}$, where $\mathbf{b}$ is the horizontal vector to that point, which fades to zero right above it, where the bearing is undefined. Soft constraints bound the height above the water, the horizontal and vertical speeds and the tilt, the angle between the thrust and the vertical,

$$\begin{aligned} & z \ge z_{min}, \qquad \left\|[\dot{x},\dot{y}]^\top\right\| \le v_{xy,max}, \\ & |\dot{z}| \le v_{z,max}, \quad \cos\phi\cos\theta \ge \cos\vartheta_{max},\end{aligned} \tag{10}$$

and the tilt is limited both on the attitude and on its command, so that the setpoint sent to the autopilot also respects it. The tether adds two more. The separation is limited, $d \le d_{max}$, and so is the rate at which it changes, the projection of the relative velocity $\dot{\mathbf{r}}$ of the winch onto the chord,

$$|\dot{d}| = \frac{|\mathbf{r}^\top\dot{\mathbf{r}}|}{d} \le \dot{d}_{max}, \tag{11}$$

which may not exceed the speed at which the winch spools the tether. Only the radial motion is restricted, so that the UAV remains free to move around the winch at a constant distance.

Disturbance observer. A force that the model does not describe, such as the wind, makes the prediction drift from the true motion and leaves a steady offset. Following the disturbance-model approach to offset-free predictive control [15], [16], an external force $\hat{\mathbf{F}}_{\text{ext}}$, held constant over the horizon, is added to (3) and estimated at every instant from the one-step prediction error of the velocity,

$$\hat{\mathbf{F}}_{\text{ext},k} = \text{sat}_{\bar{F}}\Big(\hat{\mathbf{F}}_{\text{ext},k-1} + \beta\frac{m}{T_s}\big(\mathbf{v}_k - \hat{\mathbf{v}}_{k|k-1}\big)\Big), \tag{12}$$

where T_s is the sampling time, the gain β sets how quickly the estimate follows the force, and the saturation $\bar{F}$ bounds it.

IV. Implementation

A. Simulation Environment

The system is simulated in Gazebo [17], with the hydrodynamics and the waves of the Virtual RobotX environment [18]. The USV is the WAM-V, a catamaran whose deck carries the winch and the landing pad, and which heaves, rolls and pitches on the waves. Unless stated otherwise, every test takes place in a moderate sea, with a significant wave height of 0.6 m. The

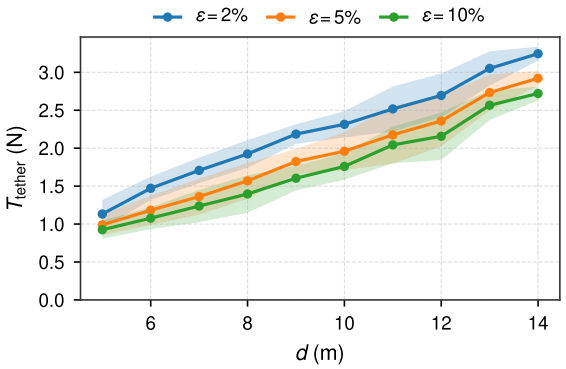


Fig. 3. Tension of the tether at the UAV in the simulator as a function of the distance between the vehicles, for three slack ratios.

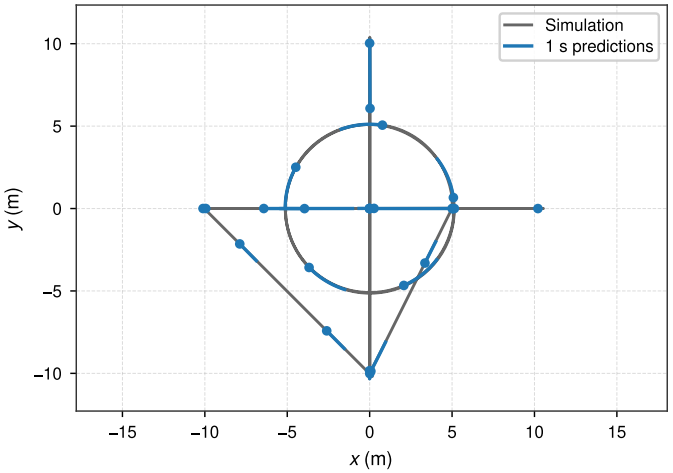


Fig. 4. Flight of the UAV in the simulator, seen from above, and predictions of 1 s of its model, each started from the state of the simulator.

UAV is the x500, a quadrotor flown by PX4 in software-in-the-loop, so that the NMPC drives it through the same autopilot as on the real vehicle. The tether is simulated with MoorDyn [7] as a chain of lumped masses with its own weight, elasticity and drag, so that it pulls on both vehicles through its own dynamics rather than through the model of the controllers. A virtual winch on the deck adjusts its length at up to 2 m/s to keep a slack of $\varepsilon_0 = 5\,\%$, and close to the deck it can instead pull the UAV towards the pad with a force of 4 N. With the slack held constant, the tension that MoorDyn computes at the UAV grows almost linearly with the distance between the vehicles, as shown in Fig. 3, since the weight of the tether that the UAV supports grows with it, and a larger slack lowers it, since a tether that sags more needs less tension to hold its own weight. Wind acts on the UAV as a horizontal drag force with gusts.

B. Model Validation

Each model of Section III was compared with the simulator in the way its controller uses it, over predictions as long as its horizon, started from the state of the simulator and driven by the recorded commands, as illustrated for the UAV in Fig. 4, with the errors, as the root-mean-square error (RMSE) and the RMSE normalised by the range of each signal (NRMSE), listed in Table I. Over its horizon, the model of the WAM-V predicts its position within a few centimetres, and that of the x500 within about one percent of the distance it travels, an accuracy that relies on the response of the attitude, without which the errors of the roll and the pitch grow to about 5 %. The tether model was compared with two simpler models that keep its tension but change its direction, a taut model, in which the force points along the chord, and a vertical model, in which the tether hangs from the UAV as a vertical load. The catenary model reduces their error by about three quarters, to 1.2 % of the weight of the UAV, and, from the positions of the two vehicles alone, follows the tension that the simulator computes from the full dynamics of the tether, as shown in Fig. 5.

TABLE I
ERRORS OF THE MODELS AGAINST THE SIMULATOR.

Model	Quantity	RMSE	NRMSE
WAM-V, 2 s	position	1.4 cm	0.37 %
	heading	0.16°	0.33 %
x500, 1 s	position	10.8 cm	1.34 %
	velocity	0.17 m/s	2.02 %
	roll, pitch	0.63°, 0.40°	1.32, 0.89 %
Tether force	catenary	0.25 N	10.8 %
	taut	0.90 N	38.9 %
	vertical	1.03 N	44.7 %

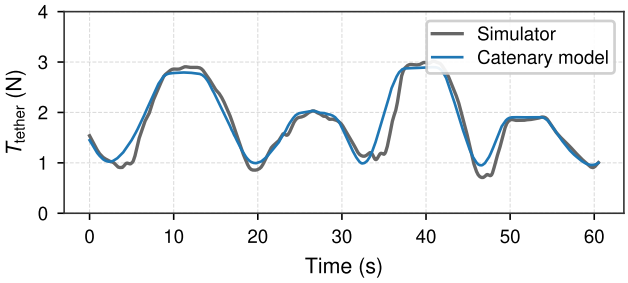


Fig. 5. Tension of the tether at the UAV, computed by MoorDyn in the simulator and by the catenary model of (5)–(6).

C. Controller Implementation

Both controllers are written in CasADi [19], from which acados [20] generates a dedicated solver in C, compiled before operation, and the models are integrated with the fourth-order Runge–Kutta method. The horizons follow the dynamics of each vehicle. The USV NMPC runs at 10 Hz over 2 s, long enough to see the slow response of the USV to its thrusters. The roll and the pitch of the UAV respond to a new command within about 0.2 s, so the UAV NMPC runs at 50 Hz, computing about ten commands while the vehicle responds to one, over a horizon of 1 s.

The USV solver iterates until convergence, which takes on average 1.1 ms and at most 3.7 ms of the 100 ms available. Iterating until convergence, the UAV solver exceeded its 20 ms in about 6 % of the instants close to its limits, so it performs a single iteration per instant instead, the real-time iteration [21], as shown in Fig. 6, which takes on average 2.2 ms and at most 4.7 ms, and whose commands differ from the converged ones

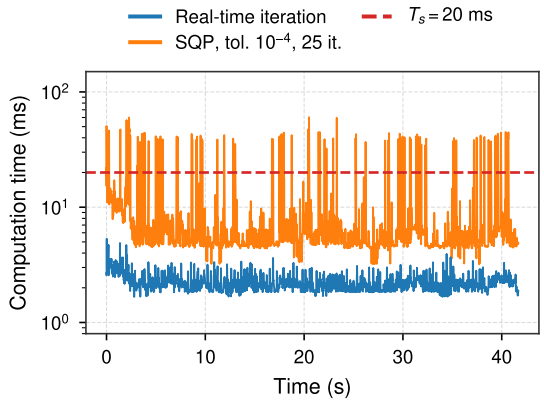


Fig. 6. Computation time of the UAV solver close to its limits, iterating until convergence and with a single iteration per instant, against the sampling time.

by less than $0.7°$.

Both controllers limit the separation between the vehicles to $d_{max} = 15$ m, raised in the missions that need a longer reach, and the UAV limits the rate at which it changes to the speed of the winch, $\dot{d}_{max} = 2$ m/s. The disturbance observer uses a gain $\beta = 0.03$, so that its estimate settles in about 0.7 s, slower than the response of the attitude but faster than the gusts of the wind, and a bound of 8 N, well above the strongest wind tested.

D. Manoeuvres and Missions

The controllers follow references, but do not decide where each vehicle should go. This is the role of the manoeuvres, which give the reference of one vehicle over the horizon of its controller at every instant. The UAV can take off, fly in formation with the USV, orbit it or a point in the world, follow a target and land, while the USV can hold its position, sail a route, follow a target, or support the UAV, sailing below it at a safe distance from a structure that the UAV circles, so that the tether does not wrap around the structure. The references are simple geometric curves, such as lines and circles, and often a single point. Figure 7 shows the manoeuvres of the UAV as flown in simulation, the takeoff and the landing through the height of the UAV above the winch, and the others from above.

The takeoff and the landing are the manoeuvres in which the UAV is closest to the deck and moves with it, on a pad of 0.40 m by 0.40 m placed directly above the winch. To keep the waves from tipping the UAV off the deck, the winch pulls it towards the pad from the start of the takeoff until it is clear of the deck. Its controller does not include this pull in its model, and the vertical component of the disturbance estimate is held while the pull acts, so that the controller does not cancel it. The takeoff therefore starts with a vertical climb off the pad, so that the UAV does not drift over the edge of the deck at low height. In the landing, the UAV first waits above the pad until it is centred and moving with it, then descends with the winch pulling it, and ends with a slow approach towards a point below the deck, so that it touches down even if the deck falls away. If it is not centred within 40 s, the landing is aborted and the USV starts the recovery of the UAV.

A mission chains the manoeuvres of both vehicles, each ending with an event, such as the detection of a target or the end of a manoeuvre of the other vehicle. Three missions were tested. In the inspection, the UAV circles a tower while the USV supports it. In the patrol, the UAV orbits the USV until it detects a target, which both vehicles then follow. In the recovery, the UAV is held at a point it cannot leave, while the USV sails away, waits at the reach of the tether and returns to collect it.

E. Software

The models, the controllers and the manoeuvres run as independent nodes of the Robot Operating System (ROS) 2 [22], connected to the simulator and to the autopilot as shown in Fig. 8, so that the controllers do not depend on whether their data comes from the simulation or from the real vehicles. An operator launches, records and aborts the missions from a web interface, the Mission Control Centre, which shows the two vehicles, the tether and their references on a map, so that every mission is run and recorded in the same way. The code of the system is publicly available[1].

V. Results

Evaluated all together, the limits of each vehicle, the coupling through the tether and the disturbances of the sea would not reveal which of them causes a given behaviour. The evaluation therefore introduces them one at a time. It begins with each vehicle alone, then exposes the landing on the moving deck, the manoeuvre most sensitive to the sea, to higher waves and to wind, and finally couples the two vehicles through the tether in complete missions.

A. Trajectory Tracking

Each controller was first given a reference within the limits of its vehicle. The USV, with the UAV resting on its deck, sails a figure-eight of two circles of 12 m radius, turned in opposite directions, as shown in Fig. 9, and follows it with an RMSE of 5.5 cm, about one percent of its length, and a heading error below one degree. The UAV, tethered to the USV, which holds its position, flies a closed trajectory around it, made of two fast arcs at a constant separation joined by two slow turns, as shown in Fig. 10, so that its speed and the force of the tether vary along each lap. As a baseline, the same trajectory was flown by the position controller supplied with PX4, which acts on the error between the current point of the trajectory and the position of the UAV, whereas the NMPC sees the trajectory one second ahead and predicts the response of the UAV and of the tether. Both command nearly the same tilt and thrust, well within the limits of the UAV, but the error of the NMPC is more than four times smaller, as listed in Table II. The error of PX4 grows on the fast arcs and shrinks on the slow turns, as shown in Fig. 11, since it trails the reference by about 80 ms, and a constant lag produces an error proportional to the speed. The NMPC, which anticipates the reference, shows no such lag.

[1] https://github.com/tomasgesteves1/tethered-usv-uav-nmpc

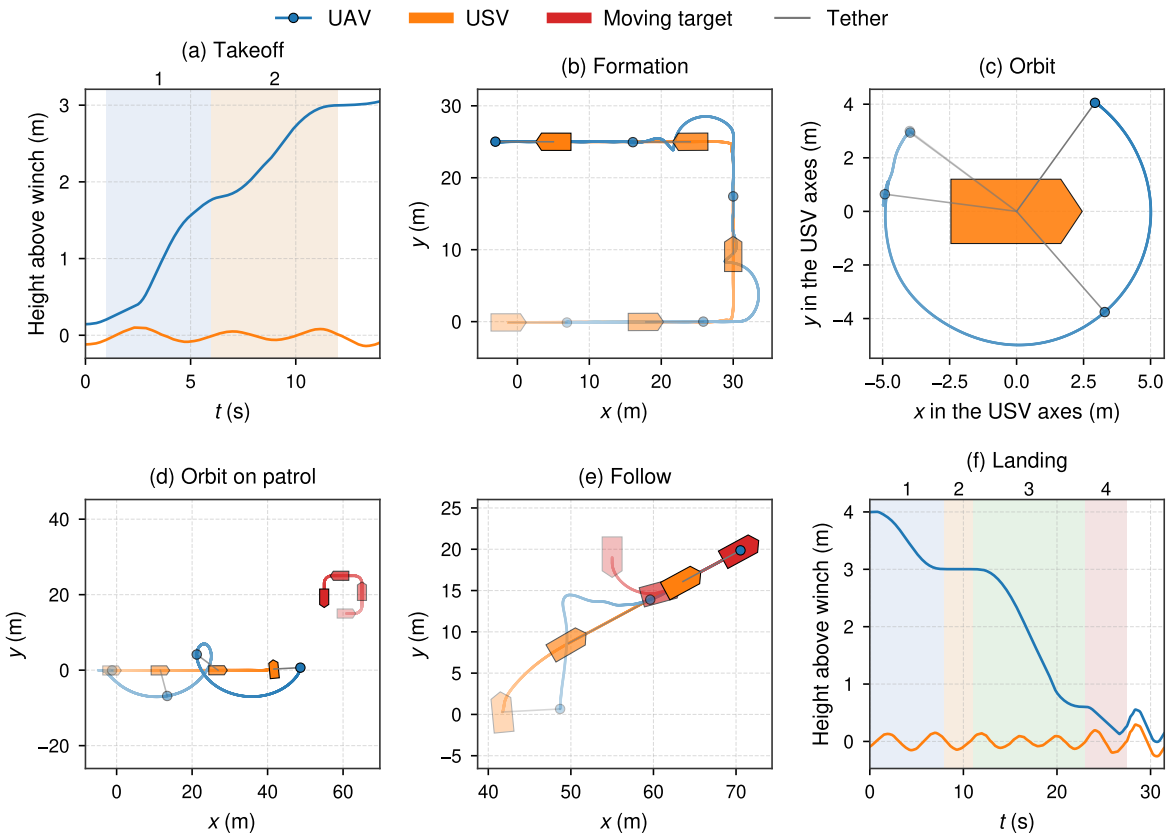


Fig. 7. Manoeuvres of the UAV in simulation.

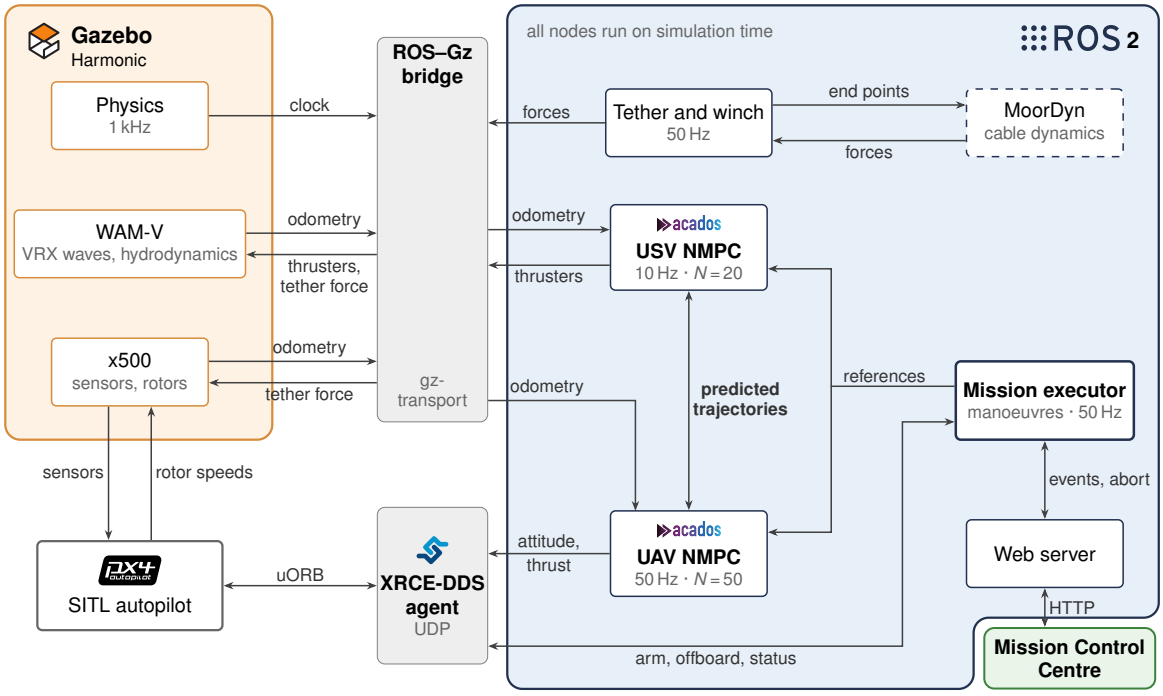


Fig. 8. Software architecture of the system.

A second reference asks each vehicle for motions it cannot perform. The UAV is given an arc of 15 m radius at 10.5 m/s, which requires a tilt of about 37° against a limit of 28.6°, a second arc at 11 m/s, beyond its speed limit of 10 m/s, and two steps in height, the second down to the surface of the water, below its minimum height of 2 m, as shown in Fig. 12.

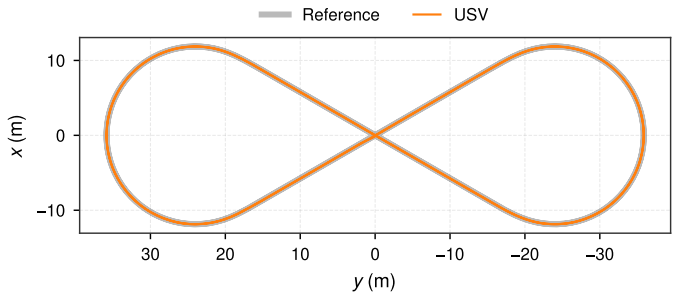


Fig. 9. Figure-eight sailed by the USV and its reference, seen from above.

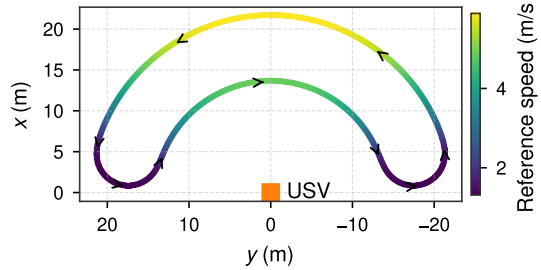


Fig. 10. Reference trajectory of the UAV around the USV, coloured by its speed, seen from above.

TABLE II
POSITION ERROR OF THE UAV (CM).

Controller	RMSE	Max.	Horizontal	Vertical
NMPC	6.5	12.8	6.4	1.0
PX4	29.9	53.8	29.9	1.8

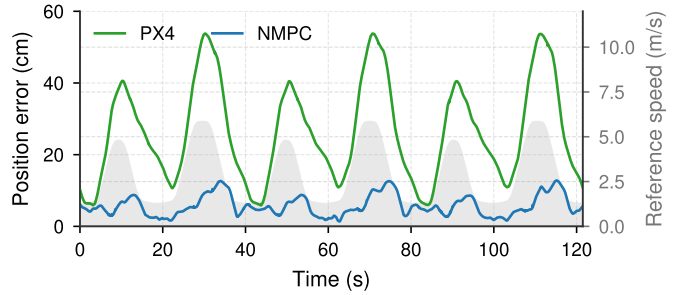


Fig. 11. Position error of the UAV over three laps, with the speed of the reference shaded.

On each arc, the NMPC drives the UAV onto the limit that makes the reference unattainable and holds it there without oscillation, falling behind the reference. Since the reference lies ahead along the arc, the NMPC steers the UAV towards it across the inside of the arc, and recovers the lag within about two seconds once the reference stops. On the way down, the minimum height enters the horizon before the UAV reaches it, and the NMPC starts to decelerate at about 3.3 m, so that the UAV comes to rest at 2 m, whereas a controller that reacted to this limit only on reaching it would cross it.

The USV is given a straight leg at 3 m/s, beyond its top speed of 2.3 m/s, ending in a sharp turn of 90°, and then a point moved 15 m backwards and 15 m to its left, which drive

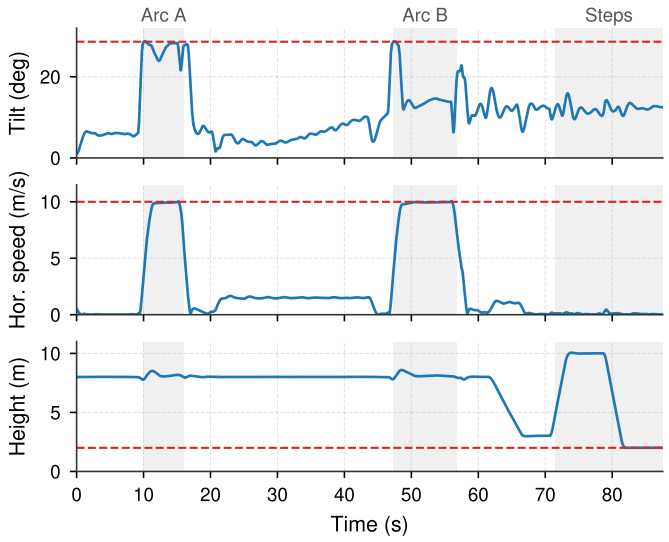


Fig. 12. Tilt, horizontal speed and height of the UAV against their limits, on a reference beyond them.

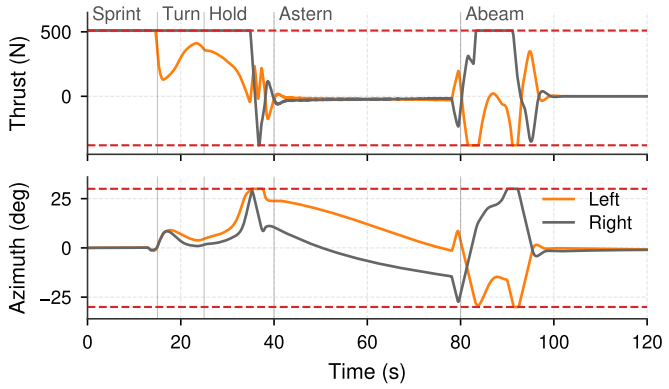


Fig. 13. Thrust and azimuth of both thrusters of the USV against their limits, on a reference beyond them.

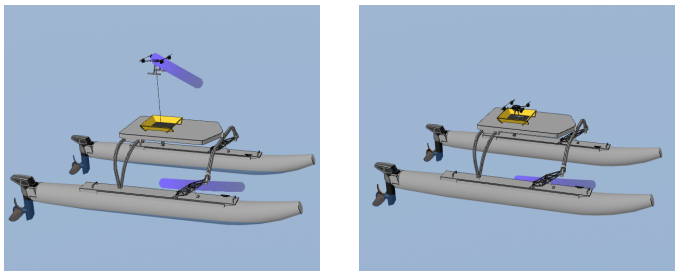
both thrusters to their limits of thrust and angle, as shown in Fig. 13. When the point moves backwards, the USV reverses in a straight line at exactly the soft lower limit of its surge speed. In both vehicles, the limits of the inputs are reached but never exceeded, and those of the states by less than 1 %, so that each controller gives up following the reference only as far as the limits of its vehicle require, and uses their full capability in doing so.

B. *Landing on the Moving Deck*

The landing closes every mission, and is the only manoeuvre that brings the UAV into contact with a deck that moves with the waves. In each run, the UAV takes off while the USV starts a route at 0.75 m/s, moves away from it and lands back on the pad, as shown in Fig. 14. A landing is judged by how well the UAV is centred above the pad when the descent starts, how close to the centre of the pad it touches down, how fast it meets the deck, and how long the landing takes, as listed in Table III.

TABLE III
LANDING UNDER WAVES AND WIND: CENTRING BEFORE THE DESCENT, DISTANCE OF THE TOUCHDOWN FROM THE CENTRE OF THE PAD, SPEED OF THE CONTACT AND DURATION.

	Centr. (cm)	**Touchd.** (cm)	**Contact** (m/s)	**Dur.** (s)
No waves	2.7	0	0.15	26.3
Waves 0.6 m	3.8	2	0.26	25.2
Waves 1.2 m	1.3	4	0.39	26.9
Wind, without / with the observer				
5 m/s	18.3 / 4.3	– / 1	– / 0.26	– / 27.7
8 m/s	44.1 / 6.4	– / 2	– / 0.25	– / 26.1



(a) Descent pulled by the winch. (b) Touchdown on the pad.

Fig. 14. Landing in the simulator, with the USV sailing at 0.75 m/s.

Without waves, each phase of the landing completes as intended. The UAV is centred within 3 cm of the winch, descends with the winch pulling it and meets the pad at 0.15 m/s, less than 1 cm from its centre, 26 s after the landing started. The landing was then repeated in seas with significant wave heights of 0.6 and 1.2 m. Since the descent is referenced to the mean height of the winch, and not to the heaving deck, the UAV descends as it does without waves, touches down within 4 cm of the centre of the pad, which is 40 cm wide, and completes the landing in the same time. What the waves change is the contact. Since the final approach descends slowly towards a point below the deck, the speed of the contact is set by the motion of the deck, and grows to 0.39 m/s in the higher sea, still below the default landing speed of PX4, of 0.7 m/s [23].

The wind was applied together with the waves of 0.6 m, at 5 and 8 m/s, the stronger producing a mean force of 2.3 N, about a ninth of the weight of the UAV. Without the disturbance observer, the wind holds the UAV away from the pad, as shown in Fig. 15, and it is never centred above it, so that the landing is aborted and the USV starts its recovery. With the observer, the UAV is centred soon after it arrives above the pad, and lands within 2 cm of its centre in the same time as without wind.

C. Missions

In the missions, both controllers operate at the same time, coupled by the tether, each executing the manoeuvres of its vehicle from the takeoff of the UAV to its landing on the deck.

Inspection. The UAV circles a tower standing in the water, 43 m high, at 13 m from its centre, in three laps at heights of 12, 27 and 42 m, pointed at the tower, so that a camera on

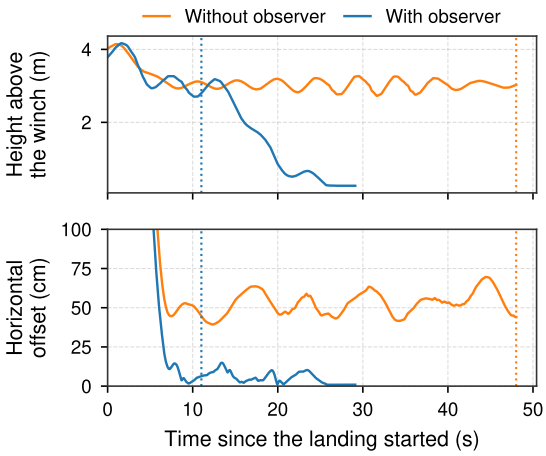
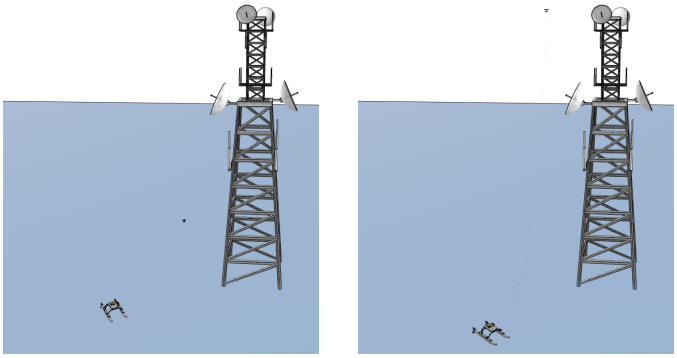


Fig. 15. Landing in a wind of 8 m/s, with and without the disturbance observer.



(a) First lap, 12 m. (b) Third lap, 42 m.

Fig. 16. Inspection of the tower in the simulator, on the lowest and the highest lap.

board covers it from bottom to top, as shown in Fig. 16. To reach the top, the maximum separation is raised to $d_{max} = 45$ m. The USV supports the UAV from the water, following it around each lap at the same angle about the tower and 12 m further out, as shown in Fig. 17, so that the tether runs outwards from the tower on every lap without crossing it. With the longer tether, its tension more than doubles from the lowest lap to the highest, yet the position error of the UAV remains of about 2 cm on all three, as shown in Fig. 18, so that the two vehicles survey the tower at a constant distance, which neither could do alone.

Patrol. The UAV acts as a lookout for the USV, orbiting it 15 m above the winch, with $d_{max} = 20$ m, while the USV sails near a zone patrolled by a target, a second WAM-V that flees at 2 m/s when either vehicle comes within 15 m. The target is detected at the edge of the view of a downward camera, beyond the reach of the tether, as shown in Fig. 19. The UAV flies towards it until the separation reaches d_{max}, and remains at the limit of the tether while the USV closes the distance. Once the target is within reach, the UAV keeps above it until the end of the pursuit, so that it never leaves the view of the camera. Throughout the mission, the UAV keeps within both limits of the tether, as shown in Fig. 20. Besides the separation

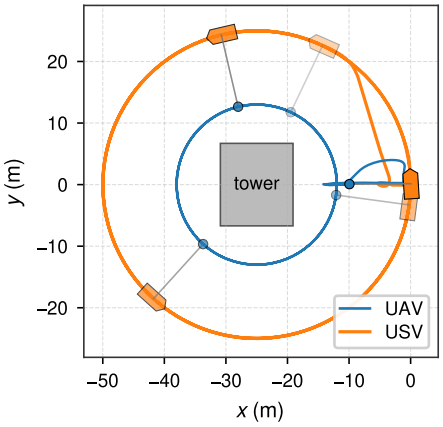


Fig. 17. Paths of both vehicles around the tower in the inspection, seen from above.

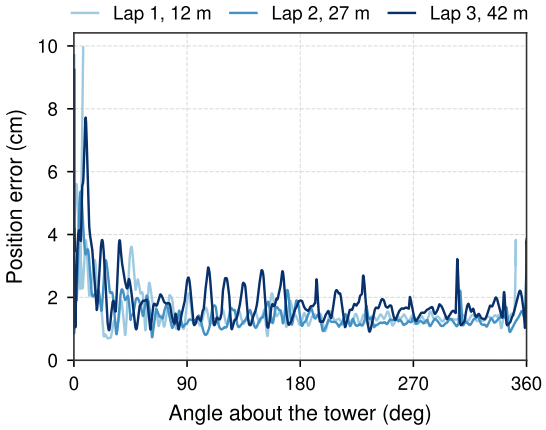


Fig. 18. Position error of the UAV along each lap of the inspection.

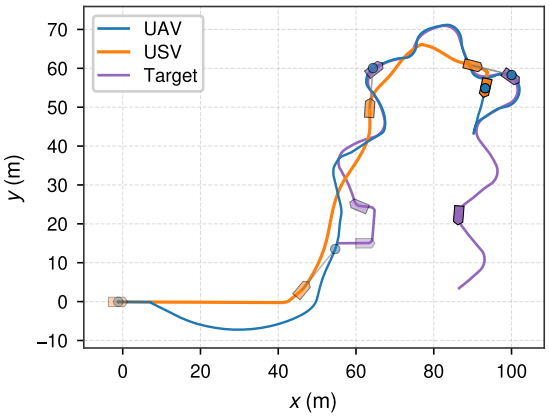


Fig. 19. Paths of both vehicles and of the target in the patrol.

at d_{max}, its rate reaches the speed of the winch as the UAV climbs to its orbit, sets off after the target and descends to land, and the two exceed their soft limits by at most $0.1\,\%$ and $6\,\%$, and the UAV then lands $1\,\text{cm}$ from the centre of the pad.

Recovery. The last mission represents the rescue of a UAV that cannot follow the USV, for instance against a wind it cannot overcome. The UAV is held at a point fixed in the

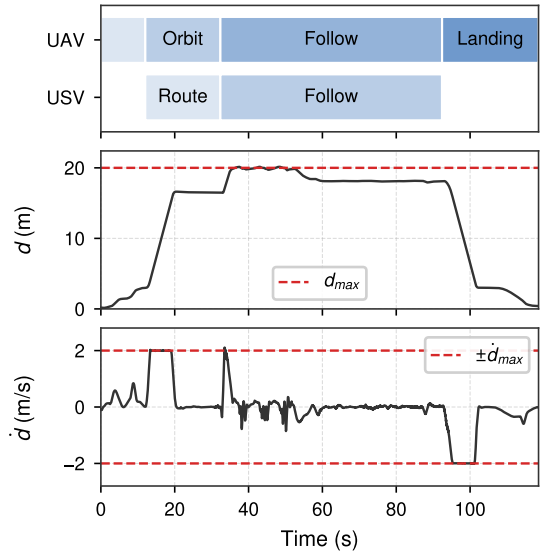


Fig. 20. Manoeuvres, separation d and its rate of change $\dot{d}$ in the patrol.

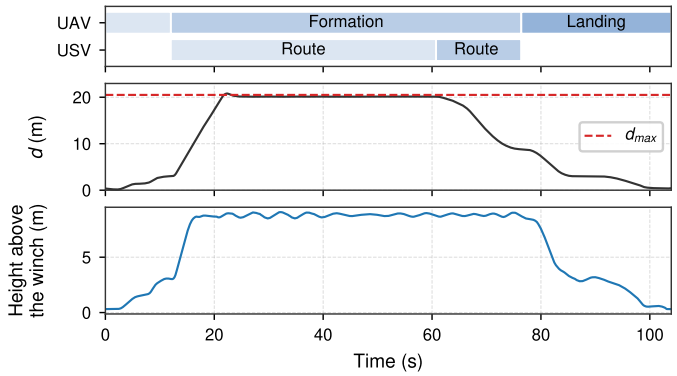


Fig. 21. Manoeuvres, separation d and height of the UAV in the recovery.

world, without giving way to the tether, while the USV is sent along a route of $60\,\text{m}$, about three times the maximum separation of $d_{max} = 20.5\,\text{m}$. The USV sets off at its top speed, brakes as the separation predicted over its horizon approaches d_{max}, and waits at the reach of the tether, as shown in Fig. 21, which it exceeds by less than $0.5\,\text{m}$ while braking, since it sees the UAV only over the shorter horizon of the UAV. Once the wait is over, the USV returns below the UAV, which lands $3\,\text{cm}$ from the centre of the pad. When the UAV cannot follow, the USV therefore moves no further than the tether allows, and can return to collect it.

VI. Conclusions

This work set out to operate a USV and a UAV connected by a tether as a single system at sea, each vehicle within its own limits and both within the reach of the tether. Models of both vehicles, derived from their physics and validated against the simulator without any fitting, predict their motion over the horizons of the controllers with errors of a few percent, and a closed-form model of a slack tether that accounts for its weight reduces the error of its force by about three quarters relative to a taut tether or a hanging load. On these models,

each vehicle is driven by its own NMPC, and the two exchange the trajectories they predict, so that each keeps the separation within the reach of the tether while anticipating where the other vehicle is going.

By anticipating the motion instead of reacting to the error, the controller of the UAV tracks a trajectory with an error more than four times smaller than that of the position controller of PX4, and, when a reference demands more than a vehicle can deliver, each controller gives up following it only as far as the limits require, so that the manoeuvres can remain simple. The UAV lands on the moving deck a few centimetres from the centre of the pad in every sea tested, and, with the disturbance observer, also under wind, which otherwise forces the landing to be aborted. Coupled by the tether, the two vehicles complete an inspection, a patrol and a recovery as a single system, and in the inspection the tether becomes not a constraint to be avoided, but a means to survey a tower that neither vehicle could survey alone.

The system was evaluated in simulation only, and the next step is to test it with the real vehicles, where the state is estimated from noisy sensors, the communication between the vehicles has delays, and the models must be validated again. A prediction of the motion of the deck could time the touchdown to when the deck moves down, and the takeoff remains to be tested under wind. Finally, missions that branch, for instance when a target is lost, and obstacles added to both NMPCs as constraints on the vehicles and on the tether, would bring the system closer to real operations.

Declaration of Artificial Intelligence Tools Use

I declare that generative Artificial Intelligence (AI) and AI-assisted tools were used in this work. Nevertheless, I take full responsibility for the content and integrity of this work. In compliance with the Guide for the Responsible Use of Artificial Intelligence at Técnico, I declare that the following tools were used:

1) As a coding assistant: Anthropic. 2026. Claude Code [AI coding assistant]. https://claude.com/claude-code
2) As a coding assistant: Google. 2026. Google Antigravity with Gemini [AI coding assistant]. https://antigravity.google
3) For linguistic revision of the text: Anthropic. 2026. Claude [Large language model]. https://claude.ai

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